

Network Intrusion Detection Technique

Problem Statement Document — Phase 1, Week 1

Project Area: Cybersecurity / Network Security / Machine Learning

Primary Focus: Detection and classification of malicious network activity using an intrusion detection approach

1. Background

Network intrusion is a major cybersecurity threat in which attackers attempt to gain unauthorized access to, disrupt, misuse, or steal information from computer networks. Common examples include denial-of-service attacks, port scanning, brute-force attempts, malware-related traffic, and unauthorized network access. A Network Intrusion Detection System (NIDS) monitors network traffic and identifies activities that may indicate an attack or security violation.

2. Problem Context

Traditional intrusion detection approaches commonly use predefined signatures, rules, blacklists, and manually created security policies. These methods can be effective for known attacks, but they may struggle with new, modified, or previously unseen attack patterns. Network traffic can also be large and continuously changing, making manual monitoring difficult. Machine-learning-based approaches can therefore be investigated to learn patterns from normal and malicious network traffic and assist in automated detection.

3. Core Problem Statement

Existing network intrusion detection approaches may achieve good performance on known datasets, but their effectiveness can be affected by changing attack patterns, imbalanced datasets, noisy network traffic, feature selection, false alarms, and limited generalization to unseen traffic. Therefore, there is a need to develop and evaluate a Network Intrusion Detection Technique that can extract meaningful network-traffic characteristics, classify traffic as normal or malicious, compare suitable detection algorithms, and evaluate how reliably the system identifies different types of intrusions.

4. Research Problem

The research problem is not simply to label network traffic as normal or attack traffic. The project should investigate whether machine-learning techniques can provide reliable intrusion detection while maintaining acceptable performance on previously unseen network traffic.

5. Proposed Problem-Solving Direction

The proposed project will investigate a network-traffic-based detection pipeline. Network traffic records or flow data will be collected from a suitable dataset, cleaned and preprocessed, transformed into measurable features, and passed to trained classification models. The system will produce a classification indicating whether the traffic is normal or potentially intrusive. Multiple algorithms may be compared to identify a suitable model for the project.

6. Project Objectives

1. Study existing Network Intrusion Detection System (NIDS) approaches through selected research literature.
2. Understand common network attacks and their traffic characteristics.
3. Prepare and preprocess a suitable normal and attack-traffic dataset.
4. Identify important network features useful for intrusion classification.
5. Implement and compare suitable machine-learning classification algorithms.
6. Evaluate the models using accuracy, precision, recall, F1-score, confusion matrix and other appropriate metrics.
7. Investigate model performance on previously unseen or independently separated data.
8. Develop a usable demonstration that can classify network traffic and present the detection result.

7. Scope of the Problem

In scope: network-traffic analysis, dataset preparation, preprocessing, feature extraction/selection, supervised machine learning, intrusion classification, model comparison, performance evaluation, and a demonstration interface or prototype.

Initially out of scope: automatic attack execution, unauthorized network testing, full enterprise-scale network deployment, guaranteed detection of every attack, and automated response/remediation. These may be considered as future extensions.

8. Main Challenges

Challenge	Research concern
Dataset dependency	A model may learn patterns specific to one dataset rather than general network attacks.
Generalization	Performance may decrease when the model is evaluated on different or unseen traffic.
Feature selection	Network datasets may contain many redundant, noisy, or less useful features.
Evolving attacks	Attackers continuously modify techniques and traffic patterns.
False positives and false negatives	Incorrect classification can either miss attacks or incorrectly flag legitimate traffic.
Class imbalance	Some attack categories may contain far fewer samples than normal traffic or other attacks.

9. Proposed High-Level Workflow

Network Traffic → Data Collection → Preprocessing → Feature Extraction/Selection → ML Classifier → Intrusion Prediction → Performance Evaluation → Detection Result

Note: This Phase 1 problem statement is intentionally broader than a single algorithm. The final dataset, features, machine-learning model, and evaluation methodology should be selected and justified using the literature reviewed by the project team.